ELECTIVE COURSE-5.2

URA622: MOBILE ROBOTICS				
	L	T	P	Cr
	2	0	2	3.0

Course Objective: The course will give students an opportunity to design and fabricate a mobile robotic platform and program it to apply learned theoretical concepts in practice.

Syllabus

Types of mobile robots, Applications of mobile robots, Robot locomotion, Types of locomotion, Degree of Maneuverability, Stability, Controllability,

Mobile Robot Kinematics and Dynamics: Forward and inverse kinematics, Holonomic and Nonholonomic constraints, Kinematic models of unicycle car, full simple car, and legged robots, Dynamic simulation of mobile robots.

Perception: Robot sensors, Passive/active sensors, performance measures of sensors, sensors for mobile robots like IMU, GPS, Wheel encoders, Accelerometers, vision-based sensors, Uncertainty in sensing, filtering, Sensor errors.

Localization: Map representation, Odometry position estimation, probabilistic mapping, Markov localization, Bayesian localization, Kalman localization, SLAM, Extended Kalman Filter (EKF) SLAM.

Planning and Navigation: Path planning algorithms based on A-star, Dijkstra, Rapid exploring random tree (RRT), Markov Decision Processes (MDP). Obstacle avoidance algorithms such as Bug algorithm, Vector field histogram, Dynamic window approach.

Robotics Project: Students will work on a semester long project consisting of design, fabrication, and programming a mobile robotic platform.

Laboratory Work

- Develop and validate kinematic model of mobile robots with different wheel configuration in simulation software i.e. MATLAB, ROS.
- Extracting and analysis data from various sensors such as IMU, Accelerometer, wheel encoders.
- Path planning with A*, RRT, and MDP algorithms in simulation software i.e. MATLAB, ROS.
- Develop the environment map using occupancy grid mapping in simulation software i.e. MATLAB, ROS.
- Real time implementing obstacle avoidance algorithms such as Bug algorithm, VFH on mobile robot platform.

Course Learning Outcomes (CLOs)

The students will be able to:

1. develop the kinematic model of the mobile robots
2. visualize and analyze the sensor's data

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

3. develop the localization algorithm for mobile robot *4.* implement the path planning and navigation algorithms
Text Books 1. R. Siegwart, I. R. Nourbakhsh, Introduction to Autonomous Mobile Robots, The MIT Press, 2011. 2. Peter Corke, Robotics, Vision and Control: Fundamental Algorithms in MATLAB, Springer Tracts in Advanced Robotics, 2011. 3. S. M. LaValle, Planning Algorithms, Cambridge University Press, 2006. (Available online http://planning.cs.uiuc.edu/) 4. Thrun, S., Burgard,W., and Fox, D., Probabilistic Robotics. MIT Press, Cambridge, MA, 2005.
Reference Books 1. Melgar, E. R., Diez, C. C., Arduino and Kinect Projects: Design, Build, Blow Their Minds, 2012. 2. H. Choset, K. M. Lynch, S. Hutchinson, G. Kantor, W. Burgard, L. E. Kavraki, and S. Thrun, Principles of Robot Motion: Theory, Algorithms and Implementations, PHI Ltd., 2005. 3. Mobile Robot Localization and Map Building,José A. Castellanos, Juan D. Tardós, Springer 2012.

Evaluation Scheme:

SI. No	Evaluation Elements	Weightage (%)
1	MST	35
2	EST	35
3	Sessional (Assignments/ Projects/ Quiz/ Lab evaluations)	30